Optimal Enhancement Control of Static Voltage Stability Margin for AC/DC Power System With Renewables Considering Control Mode Switching of DC Converter Stations

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***Abstract*—In an AC/DC power system with high penetration of renewable energy stations (RESs), the injected bus power has high random fluctuation, which brings great challenges to ensuring the sufficient static voltage stability margin (SVSM) of the system. In this paper, considering the impact of uncertain RES output and DC converter stations' control mode switching with the load growth, an optimal SVSM enhancement control model for AC/DC power system is proposed. In the model, the SVSM of the system under uncertain RES power is enhanced by regulating the power output and terminal voltage of generators and the parameters of multiple control modes of DC converter stations. By proposing a triangular convex hull relaxation method and using the uncertainty-aware model (UaM) method and the Karush–Kuhn–Tucker condition of inner-layer optimization model, the proposed bi-layer optimal SVSM enhancement control model is transformed into a single-layer nonlinear programming model, which can be directly solved by the CONOPT solver with high efficiency. Case study on the modified 39-bus AC/DC power system demonstrates that the obtained SVSM enhancement control scheme can accurately consider the influence of DC converter stations' control mode switching with the load growth, identify the influence of uncertain injected bus power on the system SVSM, and enhance the SVSM to satisfy the required secure operation level.**

***Index Terms*—AC/DC power system, control mode switching, improved relaxed LASSO, Karush–Kuhn–Tucker condition, optimal enhancement control, static voltage stability margin.**

Received 6 May 2024; revised 17 August 2024; accepted 4 October 2024. Date of publication 17 January 2025; date of current version 4 April 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 51977080, in part by the Natural Science Foundation of Guangdong Province under Grant 2023A1515240075, and in part by the U.S. National Science Foundation under Grant #2124849. Paper 2024-CDPA-0543.R1, approved for publication in the IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS by the Convergence of Data-driven and Physics-based Approaches in Power System Analysis, Optimization and Control of the IEEE Industry Applications Society. *(Corresponding author: Shunjiang Lin.)*

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Color versions of one or more figures in this article are available at https://doi.org/10.1109/TIA.2025.3530868.

Digital Object Identifier 10.1109/TIA.2025.3530868

I. INTRODUCTION

STATIC voltage stability margin (SVSM) is a commonly used index for reflecting the voltage stability degree of the current operating point of a power system, which is useful for guiding the secure operation of the power system [1]. If the SVSM of a power system is small, voltage instability cases are easy to occur with the load growth in the system operation, which will result in the power outage of the system. There have been some power outage accidents related to voltage instability in actual power systems, such as the off-grid accident in Texas, the United States of America in 2022, and the repeated crossing of low voltage problems in the northern Hebei power grid of China in 2021, and so on [2], [3]. Thus, to ensure the secure operation of a power system, the SVSM is required to be maintained above a given level, such as 0.15 in the Code on security and stability for power system in China [4]. With the increasing penetration of renewable energy generation into power systems, the impact of uncertain power output of renewable energy stations (RESs) on SVSM is becoming significant [5], [6]. Meanwhile, with the rapid increase of line-commutated converter-based high voltage direct current (LCC-HVDC) and voltage source converter-based high voltage direct current (VSC-HVDC) transmission lines in power systems, they can increase the flexible regulating ability of the system to enhance the SVSM. Thus, how to coordinate various regulation measures to enhance the SVSM of AC/DC power system considering the uncertain RES power output poses great challenge [7], [8]. The optimal SVSM enhancement control is attributed to the optimal static voltage stability (SVS) control problem, which is mainly divided into two categories [9]. One is the control strategies of regulating the current operation point to prevent voltage collapse at a future moment after load fluctuation or before an accident, which is called preventive SVS control [10], [11]. The other is the control strategies to alleviate voltage instability after the occurrence of a large disturbance in the power system and a voltage collapse, called corrective SVS control [12], [13]. The optimal SVSM enhancement control in this paper belongs to the preventive SVS control. It aims to regulate the current operation point to ensure that the SVSM meet the secure operation requirements of the system considering the uncertain RES output.

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Nowadays, many scholars have studied the optimal preventive SVS control problem. In [14], an optimal reactive power reserve management scheme to improve the system SVS based on optimal power flow (OPF) model is proposed. Mahdavi et al. [15] proposes a predictive SVS control method in a power distribution system with high penetration of photovoltaic units to maintain system voltages within an appropriate bandwidth. Hao et al. [16] proposes a multi-contingency preventive control method for SVS, and comprehensively considers the SVSM requirements under multi-contingency. Wang et al. [17] proposes a preventive control strategy based on efficient power plant model of electric vehicles to maintain the SVSM above a predefined security level. However, the above literatures do not take into account the uncertainty of RES output. There are two main categories to deal with the uncertain RES output in the optimal preventive SVS control problem. One is to consider the uncertainty as a whole in the control process, and the most conservative control strategy can be obtained. In [18], the uncertain fluctuation ranges of wind farm output are described as intervals, and a multi-objective optimal SVS control model is proposed. In [19], an optimal SVSM control model considering the uncertain wind farm output is proposed, and a probabilistic distribution control method is proposed to solve the optimal SVSM control model. The other is to establish an online preventive control model to ensure that the optimal control strategy can be updated during the change of uncertain RES output. In [20], an online preventive control optimization model for on-line voltage stability of large power grids considering multiple critical emergencies is proposed. However, when the penetration of renewable energy is high, the system state varies frequently, and the required trigger time of system regulation often leads to the failure of online preventive control to meet the actual operation requirements. Hence these two categories still have limitations in practical applications under the high uncertainty environment. Thus, to propose a new method to better accommodate the uncertainties in optimal SVS control to obtain a more secure and accurate control strategy is required.

In addition, different from pure AC power systems, more extra DC control measures can be used in AC/DC power systems. Hence the SVSM can also be enhanced by optimizing the control modes and control parameters of DC converter stations. In [21], a novel control algorithm for improving SVS by using an embedded VSC-HVDC system is proposed. In [22], a coordinate strategy of control modes of inverter and transformer is proposed to improve the SVS of AC/DC systems. The above research demonstrates that regulating the control performance of LCC/VSC-HVDC is beneficial to enhance the system SVSM. Since SVSM is reflecting the maximum load growth that a power system can bear in the current operating state. In actual situation, with the load growth, the control modes of a DC converter station may switch from one to another. If the DC converter stations' control mode switching with the load growth is not considered, the obtained SVSM of AC/DC power system will be inaccurate. If the SVSM value is evaluated to be bigger than actual situation, it will bring risks to the secure operation of the system. If the SVSM value is evaluated to be smaller than actual situation, it will bring extra control costs to the system operation. Thus, this paper will study various control modes of DC converter stations and their switching with the load growth, and take them into account in the optimal SVSM enhancement control model of AC / DC power system with renewables.

This paper makes two contributions. 1) A bi-level optimization model for optimal SVSM enhancement control of AC/DC power system with renewables is proposed, in which the inner-layer SVSM calculation model includes the constraints for describing the control mode switching of DC converter stations with the load growth. It can reflect the impact of DC converter stations' control mode switching with the load growth on the SVSM of AC/DC power system. 2) A triangle convex hull relaxation method is proposed to convexify the nonlinear control mode switching constraints of DC converter stations. The improved relaxed least absolute shrinkage and selection operator (IR-LASSO) regression based convex approximation method is used to identify the approximately convex relationship between the uncertain RES output and the bus voltage in the extended power flow equation (PFE) constraints. They can improve the solution efficiency and reliability of the optimal SVSM enhancement control model.

The rest of this paper is organized as follows. Section II describes the optimal SVSM enhancement control model of AC/DC power system with renewables. Section III introduces the solution method of the optimal SVSM enhancement control model. The case study and conclusions of the modified 39-bus AC/DC hybrid power system are displayed in Sections IV and V, respectively.

II. Optimal SVSM Enhancement Control Model of AC/DC Power System With Renewables

The optimal SVSM enhancement control model of AC/DC power system with renewables considering the control mode switching of DC converter stations takes the generators' active power output and terminal voltage, and the control parameters of DC converter stations' control modes as the control variables. Through the coordinate control of various control measures in the AC/DC system, the system SVSM in ensured to meet the given secure operation requirements. The model mainly includes the objective function, the normal operation constraints of the AC/DC power system with renewables, the control mode switching constraints of DC converter stations, and the inner-layer SVSM calculation model constraint.

A. Objective Function

The objective of the optimal SVSM enhancement control model is to minimize the control cost, which is shown in (1).

$$\begin{aligned}\min \sum_{i\in S_{\mathrm{G}}}\left[c_{\mathrm{G}i}\left(P_{\mathrm{G}i0}-P_{\mathrm{G}i0\mathrm{p}}\right)^2+c_{\mathrm{V}i}\left(V_{\mathrm{G}i\mathrm{ref}}-V_{\mathrm{G}i\mathrm{refp}}\right)^2\right]\\ +\sum_{i\in S_{\mathrm{LCCR}}}c_{\mathrm{LCC}i}\Big[\left(I_{\mathrm{d}i\mathrm{ref}}-I_{\mathrm{d}i\mathrm{refp}}\right)^2+\left(\cos\alpha_{i\mathrm{ref}}\right.\\ \left.-\cos\alpha_{i\mathrm{refp}}\right)^2+\left(k_{\mathrm{L}i\mathrm{ref}}-k_{\mathrm{L}i\mathrm{refp}}\right)^2\Big]\end{aligned}$$

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$$
\begin{aligned}
&+\sum_{i\in S_{\mathrm{LCCI}}} c_{\mathrm{LCC}i}\Big[(U_{\mathrm{d}i\mathrm{ref}}-U_{\mathrm{d}i\mathrm{refp}})^2+(\cos\gamma_{i\mathrm{ref}}\\
&-\cos\gamma_{i\mathrm{refp}})^2+(I_{\mathrm{dI}i\mathrm{ref}}-I_{\mathrm{dI}i\mathrm{refp}})^2\Big]\\
&+\sum_{i\in S_{\mathrm{VSC}}} c_{\mathrm{VSC}i}\Big[(P_{\mathrm{d}i\mathrm{ref}}-P_{\mathrm{d}i\mathrm{refp}})^2+(Q_{\mathrm{d}i\mathrm{ref}}\\
&-Q_{\mathrm{d}i\mathrm{refp}})^2+(U_{\mathrm{d}i\mathrm{ref}}-U_{\mathrm{d}i\mathrm{refp}})^2\\
&+(V_{i\mathrm{ref}}-V_{i\mathrm{refp}})^2+(k_{\mathrm{V}i\mathrm{ref}}-k_{\mathrm{V}i\mathrm{refp}})^2\Big] \qquad (1)
\end{aligned}
$$

where, $P_{\mathrm{G}i0}$ and $V_{\mathrm{G}i\mathrm{ref}}$ are the initial active power output and terminal voltage' reference value of the ith generator, $P_{\mathrm{G}i0\mathrm{p}}$ and $V_{\mathrm{G}i\mathrm{refp}}$ are their values before the optimal control, and $c_{\mathrm{G}i}$ and $c_{\mathrm{V}i}$ are the corresponding unit control costs. S_{G} is the set of generator buses. S_{LCCR} and S_{LCCI} are the sets of LCC rectifier and inverter station buses. S_{VSC} is the set of VSC converter station buses. The control variables of LCC converter stations are the control parameters of each control mode, including DC voltage $U_{\mathrm{d}i\mathrm{ref}}$, DC current $I_{\mathrm{d}i\mathrm{ref}}$, $I_{\mathrm{dI}i\mathrm{ref}}$, delay trigger angle $\alpha_{i\mathrm{ref}}$, extinction angle $\gamma_{i\mathrm{ref}}$ and the slope of voltage dependent current order limiter (VDCOL) $k_{\mathrm{L}i\mathrm{ref}}$, and $U_{\mathrm{d}i\mathrm{refp}}$, $I_{\mathrm{d}i\mathrm{refp}}$, $I_{\mathrm{dI}i\mathrm{refp}}$, $\alpha_{i\mathrm{refp}}$, $\gamma_{i\mathrm{refp}}$, $k_{\mathrm{L}i\mathrm{refp}}$ are their values before the optimal control. The control variables of VSC converter stations are also the control parameters of each control mode, including DC active power $P_{\mathrm{d}i\mathrm{ref}}$, DC reactive power $Q_{\mathrm{d}i\mathrm{ref}}$, DC voltage $U_{\mathrm{d}i\mathrm{ref}}$, AC side voltage $V_{i\mathrm{ref}}$ and the slope of P_{d}-U_{d} droop control $k_{\mathrm{V}i\mathrm{ref}}$, and $P_{\mathrm{d}i\mathrm{refp}}$, $Q_{\mathrm{d}i\mathrm{refp}}$, $U_{\mathrm{d}i\mathrm{refp}}$, $V_{i\mathrm{refp}}$, $k_{\mathrm{V}i\mathrm{refp}}$ are their values before the optimal control. $c_{\mathrm{LCC}i}$ and $c_{\mathrm{VSC}i}$ are the corresponding unit control costs LCC and VSC converter station variables.

B. Operation Constraints of AC/DC Power System With RESs

Since the PFEs in the rectangular coordinate system are quadratic equations, which has better computational performance than the PFEs in the polar coordinate system in the OPF calculation. The PFE constraints are as (2)–(7). (2)–(3) are PFEs of bus i. Assuming that RESs operate in constant power factor mode. (4) is the generator's terminal voltage constraint. (5) is the bus voltage limit constraint. (6)–(7) are the constraints of generators' power output.

$$
\begin{aligned}
&P_{\mathrm{G}i0}+P_{\mathrm{RE}i}-P_{\mathrm{L}i0}-P_{\mathrm{d}i}\\
&-e_i\sum_{j=1}^{n}(G_{ij}e_j-B_{ij}f_j)-f_i\sum_{j=1}^{n}(G_{ij}f_j+B_{ij}e_j)=0 \qquad (2)\\
&Q_{\mathrm{G}i0}+P_{\mathrm{RE}i}\tan\varphi_i-Q_{\mathrm{L}i0}-Q_{\mathrm{d}i}\\
&-f_i\sum_{j=1}^{n}(G_{ij}e_j-B_{ij}f_j)+e_i\sum_{j=1}^{n}(G_{ij}f_j+B_{ij}e_j)=0 \qquad (3)\\
&e_{Gi}^2+f_{Gi}^2=V_{G i\mathrm{ref}}^2 \qquad (4)\\
&V_{i\min}^2\le e_i^2+f_i^2\le V_{i\max}^2 \qquad (5)\\
&P_{\mathrm{G}i\min}\le P_{\mathrm{G}i0}\le P_{\mathrm{G}i\max} \qquad (6)\\
&Q_{\mathrm{G}i\min}\le Q_{\mathrm{G}i0}\le Q_{\mathrm{G}i\max} \qquad (7)
\end{aligned}
$$

where, $Q_{\mathrm{G}i0}$ is the initial reactive power output of the generator at bus i. $P_{\mathrm{RE}i}$ and φ_i are the RES's active power and power factor angle at bus i. $P_{\mathrm{L}i0}$ and $Q_{\mathrm{L}i0}$ are the active and reactive load power at bus i. $P_{\mathrm{d}i}$ and $Q_{\mathrm{d}i}$ are active and reactive power absorbed by LCC/VSC-HVDC converter station from AC system. e_i and f_i are the real and imaginary parts of the voltage of bus i. G_{ij} and B_{ij} are the real and imaginary parts of the ith row and jth column element of nodal admittance matrix. n is the total number of AC buses in the AC/DC power system. $V_{i\max}$/$V_{i\min}$ is the upper/lower limit of the voltage of bus i. $P_{\mathrm{G}i\max}$/$P_{\mathrm{G}i\min}$ and $Q_{\mathrm{G}i\max}$/$Q_{Gi\min}$ are the upper/lower limits of $P_{\mathrm{G}i0}$ and $Q_{\mathrm{G}i0}$.

The operating constraints of LCC-HVDC and multiple terminal VSC-HVDC transmission systems are shown in (8) and (9), respectively. The $\pm$ sign in the 1st equation of (8) is positive/negative for rectifier/inverter station.

$$
\begin{cases}
P_{\mathrm{d}i}=\pm K_{\mathrm{p}i}U_{\mathrm{d}i}I_{\mathrm{d}i}\\
Q_{\mathrm{d}i}=K_{\mathrm{p}i}U_{\mathrm{d}i}I_{\mathrm{d}i}\tan\varphi_{\mathrm{d}i}\\
U_{\mathrm{d}i}=3\sqrt{2}/\pi\cdot V_iK_{\mathrm{d}i}\cos\xi_i-3/\pi\cdot X_{ci}I_{\mathrm{d}i}\\
\tan^2\varphi_{\mathrm{d}i}=\left(3\sqrt{2}/\pi\cdot V_iK_{\mathrm{d}i}/U_{\mathrm{d}i}\right)^2-1\\
U_{\mathrm{dR}i}=U_{\mathrm{dI}i}+I_{\mathrm{d}i}R_{\mathrm{dc}}\\
U_{\mathrm{d}i\min}\le U_{\mathrm{d}i\mathrm{ref}}\le U_{\mathrm{d}i\max},I_{\mathrm{d}i\min}\le I_{\mathrm{d}i\mathrm{ref}}\le I_{\mathrm{d}i\max}\\
I_{\mathrm{d}i\min}\le I_{\mathrm{d}i\mathrm{Iref}}\le I_{\mathrm{d}i\max},\xi_{i\min}\le\xi_{i\mathrm{ref}}\le\xi_{i\max}\\
k_{\mathrm{L}i\min}\le k_{\mathrm{L}i\mathrm{ref}}\le k_{\mathrm{L}i\max}
\end{cases} \qquad (8)
$$

where, $U_{\mathrm{d}i}$ and $I_{\mathrm{d}i}$ are DC voltage and current of converter station of bus i. $K_{\mathrm{p}i}$ is the number of HVDC line's polar pair. V_i is voltage of bus i. $\varphi_{\mathrm{d}i}$, $K_{\mathrm{d}i}$, X_{ci} and ξ_i are the power factor angle, transformer ratio, commutation reactance and control angle of the LCC converter station of bus i. For the rectifier station, $\xi_i=\alpha_i$; and for the inverter station, $\xi_i=\gamma_i$. R_{dc} is the line resistance. $U_{\mathrm{d}i\max}$/$U_{\mathrm{d}i\min}$ is the upper/lower limits of $U_{\mathrm{d}i\mathrm{ref}}$. $I_{\mathrm{d}i\max}$/$I_{\mathrm{d}i\min}$ is the upper/lower limits of $I_{\mathrm{d}i\mathrm{ref}}$, $I_{\mathrm{dI}i\mathrm{ref}}$. $\xi_{i\max}$/$\xi_{i\min}$ and $k_{\mathrm{L}i\max}$/$k_{\mathrm{L}i\min}$ are the upper/lower limits of $\xi_{i\mathrm{ref}}$ and $k_{\mathrm{L}i\mathrm{ref}}$.

$$
\begin{cases}
P_{\mathrm{d}i}-\mu_i\eta_i/\sqrt{2}\cdot V_iU_{\mathrm{d}i}Y_i\sin\delta_i=0\\
Q_{\mathrm{d}i}+\mu_i\eta_i/\sqrt{2}\cdot V_iU_{\mathrm{d}i}Y_i\cos\delta_i-V_i^2Y_i=0\\
P_{\mathrm{d}i}-U_{\mathrm{d}i}I_{\mathrm{d}i}=0\\
I_{\mathrm{d}i}-\sum_{l=1}^{m}g_{il,\mathrm{d}}U_{\mathrm{d}l}=0\\
P_{\mathrm{d}i\min}\le P_{\mathrm{d}i\mathrm{ref}}\le P_{\mathrm{d}i\max},Q_{\mathrm{d}i\min}\le Q_{\mathrm{d}i\mathrm{ref}}\le Q_{\mathrm{d}i\max}\\
U_{\mathrm{d}i\min}\le U_{\mathrm{d}i\mathrm{ref}}\le U_{\mathrm{d}i\max},V_{i\min}\le V_{i\mathrm{ref}}\le V_{i\max}\\
k_{\mathrm{V}i\min}\le k_{\mathrm{V}i\mathrm{ref}}\le k_{\mathrm{V}i\max}
\end{cases} \qquad (9)
$$

where, μ_i is the DC voltage utilization. $_i$ is the modulation ratio of VSC-HVDC converter station. Y_i and δ_i are equivalent admittance and impedance angle difference between AC bus voltage and VSC's AC side voltage. $g_{il,\mathrm{d}}$ is the conductance between DC buses i and l. $P_{\mathrm{d}i\max}$/$P_{\mathrm{d}i\min}$ and $Q_{\mathrm{d}i\max}$/$Q_{\mathrm{d}i\min}$ are the upper/lower limits of $P_{\mathrm{d}i\mathrm{ref}}$ and $Q_{\mathrm{d}i\mathrm{ref}}$. $V_{i\max}$/$V_{i\min}$ is the upper/lower limit of $V_{i\mathrm{ref}}$. $k_{\mathrm{V}i\max}$/$k_{\mathrm{V}i\min}$ is the upper/lower limit of $k_{\mathrm{V}i}$.

In addition, the system SVSM is required to be bigger than the minimum secure operation limit $\lambda_{\min}$, and $\lambda_{\min}$ is usually set as 0.15 according to the Code on security and stability for power system in China [4].

$$
\lambda_{\mathrm{m}}\ge\lambda_{\min} \qquad (10)
$$

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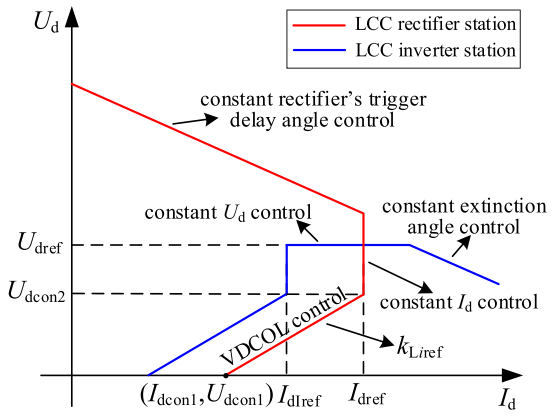


Fig. 1. Default order of the control modes of LCC converter stations.

C. Inner-Layer SVSM Calculation Model Constraints Considering DC Converter Stations' Control Mode Switching

For the variable λ_m representing the system SVSM in the constraint (10), it is necessary to establish an inner-layer SVSM calculation model to solve it. DC converter stations have multiple control modes, which can meet various operation requirement of AC/DC hybrid power systems. With the load growth, the control modes of a DC converter station may switch from one to another, and this process is required to be considered in the SVSM calculation model. Most existing publications about DC converter stations' control mode switching are as follows: in each system operation state, all possible control modes are selected and switched to the optimal control mode [23]. However, this control logic is not in line with the actual situations. In actual operation, a control mode cannot be switched to another control mode arbitrary. Taking the switching between constant DC voltage control and constant extinction angle control of the inverter station of LCC-HVDC as an example, according to the 3rd equation in (8), with the load growth, in order to keep the DC voltage U_{dIi} unchanged, it is necessary to reduce the extinction angle γ_i of the inverter station. In order to avoid the commutation failure, γ_i has a lower limit. When γ_i reaches the lower limit, the control mode of inverter station will be switched to constant minimum extinction angle control. Thus, the control mode switching of actual LCC and VSC converter stations has several relatively fixed switching processes.

References [24], [25] proposed that the control mode of a DC converter station is switched according to a certain default order. This method avoids the occurrence of unreasonable control mode switching strategy. Therefore, assuming that the LCC- and VSC- HVDC converter stations' control modes switch according to the default order. The default order of the control modes of LCC-HVDC converter stations is expressed in the U_d-I_d coordinate system [24], as shown in Fig. 1. It can be seen that according to the order of the increase of DC current I_d, the LCC-HVDC inverter side's control modes are VDCOL control, constant I_d control at I_{dIref}, constant U_d control, and constant extinction angle control sequentially. The LCC-HVDC rectifier side's control modes are VDCOL control, constant I_d control, and constant rectifier's delay trigger angle control.

It can be observed from Fig. 1 that the initial control mode of the LCC-HVDC transmission system is constant I_d control on the rectifier side and constant U_d control on the inverter side, i.e., the intersection point of two broken lines of the rectifier and inverter stations. With the load growth, the control mode of rectifier side can be switched to VDCOL control or constant rectifier's delay trigger angle control, and the control mode of inverter side can be switched to constant DC current control or constant extinction angle control. The control mode switching constraints of LCC converter stations can be described as the complementary equations of multiple control mode switching, as (11) and (12). (11) describes the control mode switching from constant U_d control to constant I_d control at I_{dIref} or constant extinction angle γ control of inverter stations, (12) describes the control mode switching from constant I_d control to constant U_d control or VDCOL control of rectifier stations.

$$\begin{cases} U_{di} = U_{di\text{ref}} - V_{d1i} - V_{d2i} \\ (I_{di} - I_{dIi\text{ref}})V_{d1i} = 0 \\ (\cos\gamma_{i\text{ref}} - \cos\gamma_i)V_{d2i} = 0 \end{cases}, i \in S_{LCCI} \tag{11}$$

$$\begin{cases} I_{di} = I_{di\text{ref}} - I_{d1i} - I_{d2i} \\ (\cos\alpha_{i\text{ref}} - \cos\alpha_i)I_{d1i} = 0 \\ [k_{Li\text{ref}} - (U_{di} - U_{dcon1})/(I_{di} - I_{dcon1})]I_{d2i} = 0 \\ k_{Li\text{ref}} = (U_{dcon2} - U_{dcon1})/(I_{dref} - I_{dcon1}) \end{cases}, i \in S_{LCCR} \tag{12}$$

where, V_{d1i} and V_{d2i} are non-negative auxiliary variables reflecting the variation of U_d caused by the control mode switching. I_{d1i} and I_{d2i} are non-negative auxiliary variables reflecting the variation of I_d caused by the control mode switching. The meaning of I_{dcon1}, U_{dcon1} and U_{dcon2} can be seen in Fig. 1.

For the VSC-HVDC transmission system, each converter station need to set two control modes related to DC side's active power P_d and reactive power Q_d during the operation [26]. In [26], the control mode switching related to Q_d with the load growth has been given, that is, the switching between constant Q_d control and constant V_i control. The control mode switching constraints are shown in (13).

$$\begin{cases} Q_{di} = Q_{di\text{ref}} - Q_{ei} \\ (V_i - V_{i\text{ref}})Q_{ei} = 0 \end{cases}, i \in S_{VSC} \tag{13}$$

where, Q_{ei} is non-negative auxiliary variable reflecting the variation of Q_{di} caused by the control mode switching.

For the control mode switching related to P_d, the control mode switching relationship is shown in Fig. 2. It can be seen that with the change of P_d, the control mode of VSC inverter station switches between constant P_d control and constant U_d control, and the control mode of VSC rectifier station switches between constant U_d control and P_d-U_d droop control. Similarly, the VSC converter stations' control mode switching constraints related to P_d with the load growth can be described as the complementary equations that are shown in (14)–(15). In the formulae, $P_{di\text{ref}}$ is $P_{dRi\text{ref}}$/$P_{dIi\text{ref}}$ for VSC rectifier/inverter

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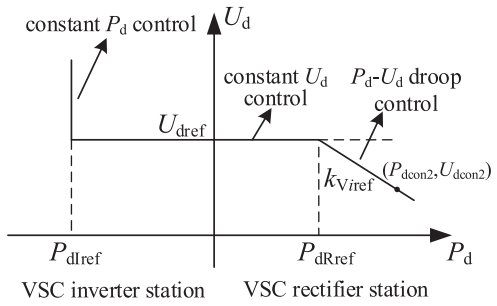


Fig. 2. Control modes of VSC converter stations.

station as shown in Fig. 2.

$$\begin{cases} U_{di} = U_{di\text{ref}} + U_{e1i} \\ (P_{di\text{ref}} - P_{di})U_{e1i} = 0 \end{cases}, i \in N_{\text{VSCI}} \tag{14}$$

$$\begin{cases} U_{di} = U_{di\text{ref}} - U_{e2i} \\ [k_{Vi\text{ref}} - (P_{di} - P_{di\text{ref}})/(U_{di} - U_{di\text{ref}})]U_{e2i} = 0 \\ k_{Vi\text{ref}} = (P_{\text{dcon2}} - P_{di\text{ref}})/(U_{\text{dcon2}} - U_{di\text{ref}}) \end{cases}, i \in N_{\text{VSCR}} \tag{15}$$

where, $U_{e1\,i}$ and $U_{e2\,i}$ are non-negative auxiliary variables reflecting the variation of U_d caused by the control mode switching. The meaning of P_{dcon2} and U_{dcon2} can be seen in Fig. 2.

For the inner-layer SVSM calculation model, the objective is to search the maximum value of load growth parameter λ. The constraints include the extended PFEs of the AC system, the operating characteristics of LCC- and VSC-HVDC transmission systems, and the control mode switching constraints of converter stations. Based on the related research on the convex approximation techniques of OPF model [27], [28], the extended PFEs in the rectangular coordinate system are quadratic equations with respect to the real and imaginary parts of each bus voltage, which is beneficial to identify the approximate convex relationship between the RES output and each bus voltage. Hence the inner-layer SVSM calculation model is shown in (16). In which $b_{\text{PL}i} = P_{\text{L}i0}$, $b_{\text{QL}i} = Q_{\text{L}i0}$, and the calculation of $b_{\text{PG}i}$ is shown in (17).

$$\lambda_{\text{m}} = \begin{cases} \max\limits_{\boldsymbol{x}} \ \lambda \\ \text{s.t.} e_i \sum_{j=1}^{n} (G_{ij}e_j - B_{ij}f_j) + f_i \sum_{j=1}^{n} (G_{ij}f_j + B_{ij}e_j) \\ \quad = P_{\text{G}i0} + \lambda b_{\text{PG}i} + P_{\text{RE}i} - (P_{\text{L}i0} + \lambda b_{\text{PL}i}) - P_{di} \\ \quad f_i \sum_{j=1}^{n} (G_{ij}e_j - B_{ij}f_j) - e_i \sum_{j=1}^{n} (G_{ij}f_j + B_{ij}e_j) \\ \quad = Q_{\text{G}i} + Q_{\text{RE}i} - (Q_{\text{L}i0} + \lambda b_{\text{QL}i}) - Q_{di} \\ (4)-(9), (11)-(15) \end{cases} \tag{16}$$

$$b_{\text{PG}i} = \frac{P_{\text{G}i\max} - P_{\text{G}i0}}{\sum_{j=1}^{N_{\text{G}}-1} (P_{\text{G}j\max} - P_{\text{G}j0})} \sum_{k=1}^{n} b_{\text{PL}k} \tag{17}$$

In summary, the optimal SVSM enhancement control model of an AC/DC power system with RESs considering the control mode switching of DC converter stations is shown in (18). Where $f(\boldsymbol{x})$ in (18) represents the objective function shown in (1). It is a bi-level optimization model, many nonlinear constraints such as PFEs and extended PFEs are included in the model, resulting in the great difficulty to solve the model directly.

$$\begin{aligned} \min \quad & f(\boldsymbol{x}) \\ \text{s.t.} \quad & \begin{cases} (2)-(10) \\ (16)-(17) \end{cases} \end{aligned} \tag{18}$$

III. Solution Methodology

To solve the above bi-level optimization model, by using the KKT condition, the inner-layer SVSM calculation model is transformed into constraints, thereby transforming the bi-level optimization model into a single-level optimization model [18]. In order to ensure that the solution obtained by the constraints after KKT condition transformation is the optimal solution of the original model, the nonlinear constraints in the inner-layer SVSM calculation model are required to be convexified at first. Thus, the original bi-layer continuous non-convex optimization model is transformed into a single-layer continuous optimization model, which can be efficiently solved by the CONOPT solver.

A. Convex Approximation of AC Extended PFEs

In the AC/DC power system with RESs, the P_{RE} on the right side of the AC extended PFEs in (16) is uncertain. For the treatment of uncertain RES power, a commonly used method is to describe the uncertain RES power P_{RE} as an interval, and transform the optimal SVSM enhancement control problem into two multi-layer optimization problems ensuring the upper and lower bounds of SVSM interval meet the secure requirements, which will increase the difficulty of solving the problem. In recent years, there is the uncertainty-aware optimization method [29], which can use the historical operation data to learn the relationship between P_{RE} and system state variables to convexly approximate the extended PFEs, and obtain a more accurate SVSM value considering the influence of uncertain P_{RE}, and it can simplify the optimal SVSM enhancement control problem considering the uncertain RES power. In actual power system operation, an accurate SVSM result considering the influence of uncertain RES power is often more practical than a SVSM interval result. In addition, the control cost of enhancing a SVSM value to meet the secure operation requirements is often smaller than that of enhancing a SVSM interval as a whole to meet the requirements. Therefore, the uncertainty aware optimization method to deal with the uncertain RES power and the convex approximation method are used to deal with the non-convex AC extended PFEs.

To represent the mapping between the injected bus power and the real and imaginary parts of bus voltage at the SVS limit point, the extended PFEs in (16) can be compactly expressed as follows:

$$g_i(\boldsymbol{x}) = \boldsymbol{x}^{\text{T}} \boldsymbol{A'}_i \boldsymbol{x} = \tilde{u}_i, i = 1, 2 \ldots, 2n \tag{19}$$

where $\boldsymbol{x} = [\boldsymbol{e}; \boldsymbol{f}]$. Since the extended PFEs are nonconvex nonlinear, the coefficient matrix $\boldsymbol{A}'_i$ in (19) is obviously indefinite. u_i is the injected power of ith bus. When $i = 1, 2, \ldots, n$, u_i is the injected active power of ith bus P_i, including the generator active power, RES active power and active load power at ith bus. When the index $i = (n+1), (n+2), \ldots, 2n$, u_i is the injected

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reactive power of (i-n)th bus Q_i, including the reactive power output of the generator, the reactive power output of RES and reactive load power at (i-n)th bus. $\tilde{u}_i$ is the predicted injected power of ith bus.

Since convex functions have the property that the Hessian matrices are positive semidefinite (PSD), the matrices of the extended PFEs in (16) need to be approximated as PSD matrices. Hence, the quadratic mappings are described as the form of (20)–(21), where $\boldsymbol{AP}\ i$, $\boldsymbol{AQ}\ i$ are PSD matrices. By relaxing "$=$" into "$\leq$", (20)–(21) can be reformulated as convex quadratic inequalities.

$$\boldsymbol{x}^{\mathrm{T}}\boldsymbol{A}_i^P\boldsymbol{x}+\boldsymbol{B}_i^P\boldsymbol{x}+c_i^P=P_i \tag{20}$$

$$\boldsymbol{x}^{\mathrm{T}}\boldsymbol{A}_i^Q\boldsymbol{x}+\boldsymbol{B}_i^Q\boldsymbol{x}+c_i^Q=Q_i \tag{21}$$

Based on [30], the training model of IR-LASSO regression is established by combining with the adjacency matrix of the system to solve the coefficients of the extended PFEs, and the convex approximation form of extended PFEs can be obtained. The training model of IR-LASSO regression is shown in (22).

$$\begin{aligned}
&\min_{\boldsymbol{A}_i,\boldsymbol{B}_i,c_i}\frac{1}{|\boldsymbol{D}|}\sum_k r^{(k)}+\phi\sum_j|a_j|\\
&\text{s.t.}\quad (\hat{\boldsymbol{x}}^{(k)})^{\mathrm{T}}\{\boldsymbol{A}_i\odot\boldsymbol{M}_i\}\hat{\boldsymbol{x}}^{(k)}+\boldsymbol{B}_i^{\mathrm{T}}\hat{\boldsymbol{x}}^{(k)}+c_i-\tilde{u}_i^{(k)}=m^{(k)}\leq 0\\
&\quad\{\boldsymbol{A}_i\odot\boldsymbol{M}_i\}\succeq 0\\
&\quad\begin{bmatrix}1 & m^{(k)}\\ m^{(k)} & r^{(k)}\end{bmatrix}\succeq 0\\
&\begin{cases}\{\boldsymbol{M}_i\}_{j,l}=1,\{\boldsymbol{M}_i\}_{j+n,l}=1,\{\boldsymbol{M}_i\}_{j,l+n}=1,\\ \quad\{\boldsymbol{M}_i\}_{j+n,l+n}=1,(j,l)\in\boldsymbol{\psi}_{jl}\\ \{\boldsymbol{M}_i\}_{j,l}=0,\{\boldsymbol{M}_i\}_{j+n,l}=0,\{\boldsymbol{M}_i\}_{j,l+n}=0,\\ \quad\{\boldsymbol{M}_i\}_{j+n,l+n}=0,(j,l)\notin\boldsymbol{\psi}_{jl}\end{cases}
\end{aligned}\tag{22}$$

where, $\boldsymbol{D}$ is a training data set. $k=1,\ \ldots,|\boldsymbol{D}|$. The penalty term in the IR-LASSO regression model is a nonnegative hyperparameter ϕ multiplied by the 1-norm of the vector $\boldsymbol{a}$. a_j is the jth element of the column vector $\boldsymbol{a}$ which is composed of all columns in $\boldsymbol{A}_i$ matrix arranged in order. $\boldsymbol{M}_i$ is an indicator function for the index of the matrix $\boldsymbol{A}_i$, which indicates that the corresponding elements in $\boldsymbol{A}_i$ are estimated as non-zero or zero. The set of the index of elements that are estimated as non-zero is denoted as $\boldsymbol{\psi}$. $\boldsymbol{A}_i\odot\boldsymbol{M}_i$ is the Hadamard product of matrices $\boldsymbol{A}_i$ and $\boldsymbol{M}_i$. $m^{(k)}$ and $r^{(k)}$ are auxiliary variables of IR-LASSO regression model.

The extended PFEs after uncertainty-aware convex approximation are shown in (23), which can identify the influence of uncertain P_{RE} on the system states simultaneously.

$$\begin{cases}(\hat{\boldsymbol{x}}^{(k)})^{\mathrm{T}}\{\boldsymbol{A}_i^P\odot\boldsymbol{M}_i\}\hat{\boldsymbol{x}}^{(k)}+(\boldsymbol{B}_i^P)^T\hat{\boldsymbol{x}}^{(k)}+c_i^P\\ \leq P_{\mathrm{G}i}+P_{\mathrm{RE}i}-(P_{\mathrm{Li0}}+\lambda b_{\mathrm{PL}i})-P_{\mathrm{d}i}\\ (\hat{\boldsymbol{x}}^{(k)})^{\mathrm{T}}\{\boldsymbol{A}_i^Q\odot\boldsymbol{M}_i\}\hat{\boldsymbol{x}}^{(k)}+(\boldsymbol{B}_i^Q)^T\hat{\boldsymbol{x}}^{(k)}+c_i^Q\\ \leq Q_{\mathrm{G}i}+Q_{\mathrm{RE}i}-(Q_{\mathrm{Li0}}+\lambda b_{\mathrm{QL}i})-Q_{\mathrm{d}i}\end{cases}\tag{23}$$

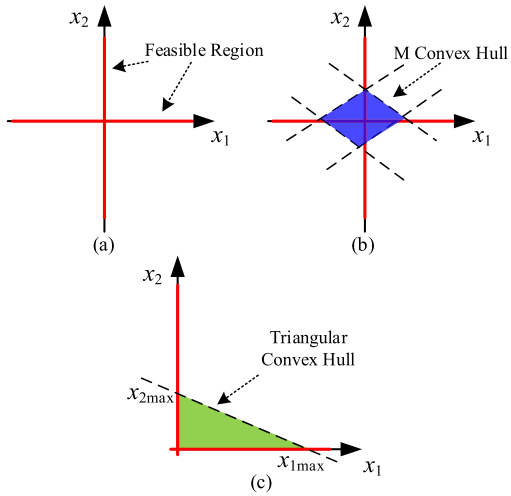


Fig. 3. Triangular convex hull relaxation method.

B. Convex Relaxation of Control Mode Switching Constraints of DC Converter Stations

For the control mode switching constraints of DC converter stations in (16), a method of constructing the triangular convex hull is proposed to relax the complementary equations with high non-convexity. Taking the 2nd equation of (11) as an example, the complementarity equation can be regarded as an equality in which two continuous variables x_1 and x_2 are multiplied and equal to 0. The feasible region of the equation in the coordinate system x_1-x_2 is as the red region shown in Fig. 3(a), it can be seen that the feasible region is a region with poor convexity. If the convex hull of the multiplication of x_1 and x_2 is constructed directly by the convex hull relaxation method and let it be 0, the newly constructed convex hull region is shown in Fig. 3(b). It can be seen that the new convex hull region adds more irrelevant regions than the original feasible region, which will result in the big relaxation gap of the optimization result. By observing the 2nd equation of (11), in order to satisfy the control mode constraint, the lower bound of I_{d} is I_{dIref}, and the auxiliary variable $V_{\mathrm{d1}i}$ also has a lower bound of 0, that is, the corresponding two continuous variables x_1, x_2 only in the first quadrant of the rectangular coordinate system. Thus, a triangular convex hull relaxation method is proposed, and an inequality constraint is added to constrain the feasible region in the triangular region of the first quadrant, as shown in Fig. 3(c). The upper bounds of x_1 and x_2 are also required to be known to obtain the relaxed feasible region, as shown in (24).

$$\begin{cases}0\leq x_1\leq x_{1\max}\\ 0\leq x_2\leq x_{2\max}\\ x_{2\max}x_1+x_{1\max}x_2\leq x_{1\max}x_{2\max}\end{cases}\tag{24}$$

The advantage of the proposed triangular convex hull relaxation method is that its relaxation gap is smaller than that of the original M convex hull [31]. In addition, the construction of triangular convex hull relaxation method does not add auxiliary variables, and the number of new relaxation constraints is

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less, which can reduce the complexity of the subsequent KKT condition transformation.

After convexification, the control mode switching constraints (11)–(15) of DC converter stations are transformed into the form of linear equalities and inequalities, as shown in (25)–(29).

$$\begin{cases} U_{di} = U_{di\text{ref}} - V_{d1i} - V_{d2i} \\ 0 \le I_{di} - I_{dIi\text{ref}} \le I_{d\max} \\ 0 \le V_{d1i} \le V_{e\max} \\ V_{e,\max}(I_{di} - I_{dIi\text{ref}}) + I_{d\max}V_{d1i} \le V_{e\max}I_{d\max} \\ 0 \le \cos\gamma_i - \cos\gamma_{i\text{ref}} \le (\cos\gamma_i)_{\max} \\ 0 \le V_{d2i} \le V_{e\max} \\ V_{e,\max}(\cos\gamma_i - \cos\gamma_{i\text{ref}}) + (\cos\gamma_i)_{\max}V_{d2i} \\ \quad \le V_{e\max}(\cos\gamma_i)_{\max} \end{cases} \tag{25}$$

$$\begin{cases} I_{di} = I_{di\text{ref}} - I_{d1i} - I_{d2i} \\ 0 \le \cos\alpha_i - \cos\alpha_{i\text{ref}} \le (\cos\alpha_i)_{\max} \\ 0 \le I_{d1i} \le I_{e\max} \\ I_{e\max}(\cos\alpha_i - \cos\alpha_{i\text{ref}}) + (\cos\alpha_i)_{\max}I_{d1i} \le I_{e\max}(\cos\alpha_i)_{\max} \\ 0 \le [(U_{di} - U_{d\text{con}1}) - k_{Li\text{ref}}(I_{di} - I_{d\text{con}1})] \le k_{1\max} \\ 0 \le I_{d2i} \le I_{e\max} \\ I_{e\max}[(U_{di} - U_{d\text{con}1}) - k_{Li\text{ref}}(I_{di} - I_{d\text{con}1})] \\ \quad + k_{1\max}I_{d2i} \le I_{e\max}k_{1\max} \end{cases} \tag{26}$$

$$\begin{cases} Q_{di} = Q_{di\text{ref}} - Q_{ei} \\ 0 \le V_i - V_{i\text{ref}} \le V_{a\max} \\ 0 \le Q_{ei} \le Q_{e\max} \\ Q_{e\max}(V_i - V_{i\text{ref}}) + V_{a\max}Q_{ei} \le Q_{e\max}V_{a\max} \end{cases} \tag{27}$$

$$\begin{cases} U_{di} = U_{di\text{ref}} + U_{e1i} \\ 0 \le P_{di\text{ref}} - P_{di} \le P_{d\max} \\ 0 \le U_{e1i} \le U_{e\max} \\ U_{e\max}(P_{di\text{ref}} - P_{di}) + P_{d\max}U_{e1i} \le U_{e\max}P_{d\max} \end{cases} \tag{28}$$

$$\begin{cases} U_{di} = U_{di\text{ref}} - U_{e2i} \\ 0 \le [(P_{di} - P_{di\text{ref}}) - k_{Vi\text{ref}}(U_{di} - U_{di\text{ref}})] \le k_{2\max} \\ 0 \le U_{e2i} \le U_{e\max} \\ U_{e\max}[(P_{di} - P_{di\text{ref}}) - k_{Vi\text{ref}}(U_{di} - U_{di\text{ref}})] \\ \quad + k_{2\max}U_{e2i} \le U_{e\max}k_{2\max} \end{cases} \tag{29}$$

where, $I_{d\max}$ is the maximum value of (I_{di}-$I_{dIi\text{ref}}$). $V_{e\max}$ is the maximum value of V_{d1i}, V_{d2i}. $(\cos\gamma_i)_{\max}$ and $(\cos\alpha_i)_{\max}$ are the maximum values of $\cos\gamma_i$ and $\cos\alpha_i$. $k_{1\max}$ is the maximum value of [(U_{di} - $U_{d\text{con}1}$) - $k_{Li\text{ref}}$($I_{di\text{ref}}$ - $I_{d\text{con}1}$)]. $V_{a\max}$ is the maximum value of (V_i-$V_{i\text{ref}}$). $Q_{e\max}$ is the maximum value of Q_{ei}. $P_{d\max}$ is the maximum value of ($P_{di\text{ref}}$-P_{di}). $U_{e\max}$ is the maximum value of U_{e1i}, U_{e2i}. $k_{2\max}$ is the maximum value of [(P_{di} - $P_{di\text{ref}}$) - $k_{Vi\text{ref}}$(U_{di} - $U_{di\text{ref}}$)].

In addition, the remaining non-convex parts in the constraints of (16) are convexified by using the convex hull relaxation and second-order cone relaxation methods. Firstly, based on the convex hull relaxation method, two basic forms of T convex hull and M convex hull relaxation are listed as (30), and the detail convex relaxation expressions are shown in (31)–(32) [30].

$$z_1 = \langle x^2 \rangle^{\mathrm{T}}, z_2 = \langle xy \rangle^{\mathrm{M}} \tag{30}$$

$$\begin{cases} x^2 - z_1 \le 0 \\ z_1 - (x_{\max} + x_{\min})x + x_{\max}x_{\min} \le 0 \end{cases} \tag{31}$$

$$\begin{cases} x_{\min}y + y_{\min}x - x_{\min}y_{\min} - z_2 \le 0 \\ x_{\max}y + y_{\max}x - x_{\max}y_{\max} - z_2 \le 0 \\ z_2 - x_{\min}y - y_{\max}x + x_{\min}y_{\max} \le 0 \\ z_2 - x_{\max}y - y_{\min}x + x_{\max}y_{\min} \le 0 \end{cases} \tag{32}$$

For the convex relaxation of the operating constraints of LCC-HVDC transmission system. The nonlinear term U_dI_d in the constraint of the DC active power P_d in (8) can be expanded by the substituting the first formulae of (25) and (26). Since the auxiliary variables representing the change of U_d and I_d are generally very small, and the quadratic term of the product of two auxiliary variables is ignored, and the linear expression of P_d can be written in (33). For the DC reactive power Q_d, a non-convex expression (34) can be obtained by combining the 2nd and 4th formulae in (8), which can be relaxed into a convex inequality (35) by using the second-order cone relaxation method. For the nonlinear term $V_i\cos\xi_i$ in (8), the M convex hull relaxation method can be used to relax it. Therefore, after convexification, the operating constraints of LCC-HVDC (8) are transformed into the form as (36).

$$\begin{aligned} P_{di} = K_{pi}[&U_{di\text{ref}}I_{di\text{ref}} - U_{di\text{ref}}(I_{d1i} + I_{d2i}) \\ &- I_{di\text{ref}}(V_{d1i} + V_{d2i})] \end{aligned} \tag{33}$$

$$(P_{di}/K_{pi})^2 + (Q_{di}/K_{pi})^2 = \left(3\sqrt{2}K_{di}\Big/\pi \cdot V_iI_{di}\right)^2 \tag{34}$$

$$\left\| \begin{matrix} P_{di}/K_{pi} \\ Q_{di}/K_{pi} \end{matrix} \right\|_2 \le 3\sqrt{2}K_{di}\Big/\pi \cdot V_iI_{di\text{ref}} \tag{35}$$

$$\begin{cases} (33), (35) \\ w_i = \langle V_i\cos\xi_i \rangle^{\mathrm{M}} \\ U_{di} = 3\sqrt{2}/\pi \cdot K_{di}w_i - 3/\pi \cdot X_{ci}I_{di} \\ U_{dRi} = U_{dIi} + I_{di}R_{dc} \end{cases} \tag{36}$$

The AC PFEs in the rectangular coordinate system don't include V_i explicitly. Combined with the T convex hull, the nonlinear constraints (4) and (5) can be convexified into the following form.

$$\begin{cases} E_{Gi} + F_{Gi} = U_{Gi\text{ref}} \\ V_{i\min}^2 \le E_i + F_i \le V_{i\max}^2 \\ E_i + F_i = U_i, E_i = \langle e_i^2 \rangle^{\mathrm{T}}, F_i = \langle f_i^2 \rangle^{\mathrm{T}}, U_i = \langle V_i^2 \rangle^{\mathrm{T}} \end{cases} \tag{37}$$

For the convex relaxation of the operating constraints of VSC-HVDC transmission system. Similarly, the expression of P_d can also be expressed as (33), and Q_d can be combined with (9) to derive its corresponding second-order cone relaxation constraint, as shown in (38). Therefore, after convexification, the operating constraints of VSC-HVDC (9) are transformed into the

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form as (39).

$$\left\| \begin{matrix} 2P_{\mathrm{d}i} \\ 2U_iY_i - 2Q_{\mathrm{d}i} \\ \mu_i U_{\mathrm{diref}} Y_i \big/ \sqrt{2} \cdot (U_i - O_i) \end{matrix} \right\|_2 \le \mu_i U_{\mathrm{diref}} Y_i \big/ \sqrt{2} \cdot (U_i + O_i) \tag{38}$$

where, $O_i = \langle \eta_i^2 \rangle^{\mathrm{T}}$. O_i is the square term of the modulation ratio $_i$ of VSC converter stations.

$$\begin{cases} (33), (38) \\ O_i = \langle \eta_i^2 \rangle^{\mathrm{T}} \end{cases} \tag{39}$$

In summary, the inner-layer SVSM calculation model after convex relaxation is shown in (40).

$$\lambda_{\mathrm{m}} = \begin{cases} \max\limits_{\boldsymbol{x}} \lambda \\ \text{s.t. } (23) \\ \quad P_{\mathrm{G}i} = P_{\mathrm{G}i0} + \lambda b_{\mathrm{PG}i} \\ \quad (6) - (7) \\ \quad (25) - (29), (36), (37), (39) \end{cases} \tag{40}$$

C. KKT Condition Transformation of the Inner-Layer SVSM Calculation Model

For the convex inner-layer SVSM calculation model, it can be transformed into its equivalent KKT condition. The convex inner-layer model (40) can be written as (41). The KKT condition of the convex inner-layer optimization model (41) is shown in (42). It can reflect the relationship between the SVSM and the control variables accurately.

$$\begin{cases} \max\limits_{\boldsymbol{x}} \lambda \\ \text{s.t.} \quad \boldsymbol{E}(\boldsymbol{x}, \lambda) = 0 \\ \quad \boldsymbol{H}(\boldsymbol{x}, \lambda) \le 0 \end{cases} \tag{41}$$

where $\boldsymbol{E}(\boldsymbol{x}, \lambda)$ and $\boldsymbol{H}(\boldsymbol{x}, \lambda)$ are the equality constraints and inequality constraints of the inner-layer model, respectively, and $\boldsymbol{x}$ is composed of state variables.

$$\begin{cases} \nabla_{\boldsymbol{x},\lambda}\lambda - \nabla_{\boldsymbol{x},\lambda}\boldsymbol{E}(\boldsymbol{x}, \lambda)\boldsymbol{y} - \nabla_{\boldsymbol{x},\lambda}\boldsymbol{H}(\boldsymbol{x}, \lambda)\boldsymbol{\mu} = \boldsymbol{0} \\ \boldsymbol{E}(\boldsymbol{x}, \lambda) = \boldsymbol{0}, \boldsymbol{y} \ne \boldsymbol{0} \\ \boldsymbol{H}(\boldsymbol{x}, \lambda) \le \boldsymbol{0}, \boldsymbol{\mu} \ge \boldsymbol{0}, \boldsymbol{\mu}\boldsymbol{H}(\boldsymbol{x}, \lambda) = \boldsymbol{0} \end{cases} \tag{42}$$

where $\boldsymbol{y}$ is the Lagrange multiplier vector of equality constraints $\boldsymbol{E}(\boldsymbol{x}, \lambda)$. μ is the Lagrange multiplier vector of inequality constraints $\boldsymbol{H}(\boldsymbol{x}, \lambda)$.

After the inner-layer SVSM calculation model is transformed into its KKT condition, the optimal SVSM enhancement control model (18) can be rewritten as (43). This model is a single-layer continuous optimization model, which can be solved by the CONOPT solver in the GAMS software.

$$\min f(x) \quad \text{s.t.} \begin{cases} (2) - (10), (17) \\ (42) \end{cases} \tag{43}$$

IV. Case Study

The proposed optimal enhancement control of SVSM model for AC/DC power system is applied to the modified IEEE-39 bus systems to demonstrate the effectiveness. An RES is integrated into the bus 9 of the system, the corresponding uncertain active output interval is [500, 1400] MW. The bus 39 is set as the swing bus. Based on the IEEE-39 bus system, the AC line 14-4 is replaced with a two-terminal LCC-HVDC line, the AC lines 11-6 and 11-12 are replaced with a three-terminal VSC-HVDC system, as shown in Fig. 4. For the two-terminal LCC-HVDC transmission line, $R_{\mathrm{dc}} = 0.02$, $X_{\mathrm{c}} = 0.1$; on the rectifier side, $K_{\mathrm{d4}} = 1.1$, and on the inverter side, $K_{\mathrm{d14}} = 1.05$; the upper and lower limits of θ_i are 15° and 8°, respectively; and the upper and lower limits of $\tan\varphi_i$ are 0.6 and 0.4, respectively. For the three-terminal VSC-HVDC transmission system, the resistance of each line is $R_{\mathrm{dc}} = 0.01$, and the equivalent admittance and impedance angle of the VSC-HVDC converter station are $Y_i = 6.661$ and $\alpha_i = 1.53$. The upper and lower limits of M_i are 1 and 0.8, respectively; the upper and lower limits of δ_i are 5° and 0°, respectively. For the LCC-HVDC transmission line, the control mode of constant $I_{\mathrm{d}i}$ and $U_{\mathrm{dR}i}$ is adopted. For the three-terminal VSC-HVDC transmission line, the control mode of constant $U_{\mathrm{d}i}$ and $Q_{\mathrm{vd}i}$ for the rectifier station and constant $I_{\mathrm{d}i}$ and $Q_{\mathrm{vd}i}$ for the inverter station is adopted. Combined with the DC line operation constraints, all the DC variables can be calculated, and they are shown in Table I. The reactive powers Q_{vd6}, Q_{vd11}, and Q_{vd12} are –0.80, –0.13, and –0.30, respectively. The used computer is a DELL OptiPlex 3010 workstation with a 3.50 GHz Intel Xeon processor E3-1270 and 32GB RAM. The used GAMS version is 24.5.3 r54322 WEX-WEI x86 64bit/MS Windows.

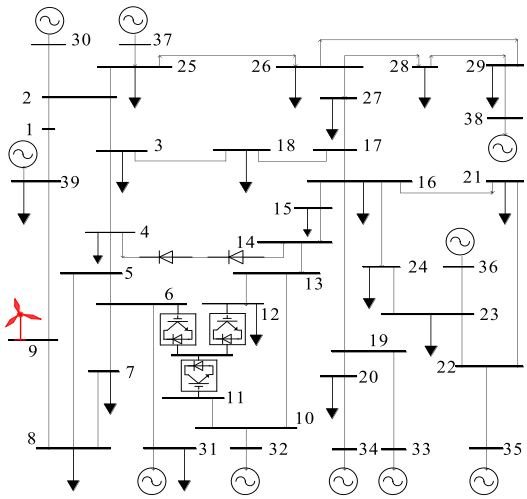


Fig. 4. Modified IEEE 39-bus AC/DC system.

TABLE I
DC Variable Values of HVDC Transmission Lines

Bus	$U_{\mathrm{d}i}$	$I_{\mathrm{d}i}$
4	1.2394	0.53
14	1.2500	0.53
6	1.9974	-0.05
11	2.0000	0.08
12	1.9978	-0.03

A. Analysis of the Convexification Methods

In order to fit the mapping between the real-time output responses and the uncertain injected bus power accurately, the training data is expected to come from the operation of a real

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TABLE II
COMPARISON OF SVSM VALUES AND CPU TIME IN DIFFERENT SCENARIOS

P_{RE} /MW	Scenario 1	CPU time/s	Scenario 2	CPU time/s	Scenario 3	CPU time/s
500	0.1451	0.207	0.1446	0.217	0.1501	0.209
600	0.1624	0.208	0.1608	0.220	0.1695	0.229
1000	0.2102	0.219	0.2071	0.226	0.2159	0.230
1300	0.1633	0.201	0.1628	0.221	0.1681	0.243
1400	0.1321	0.218	0.1319	0.223	0.1394	0.231

power system, which includes the output responses, the real-time and predicted injected bus power at each moment. The data with the above requirements is not public and difficult to obtain. As empirical alternatives, the active and reactive load curves are based on the real operation data curves and fluctuate at the range of [0.7, 1.3] around their real operation value. The reference RES power and load curves are obtained from the real 14-day RES power and total load data of 15 states in the central United States in 2023 [32]. 200 of the 336 data points obtained from the reference curves are selected as the sample set in the training model. The training process is performed in MATLAB with CVX package, and the training model (22) are solved by the MOSEK solver.

In order to demonstrate the effectiveness of the proposed convex approximation method of extended PFEs, triangular convex hull relaxation method and other used convex relaxation methods, comparison of the obtained SVSM results by solving the inner-layer model under different convexification stages is shown in Table II. Among them, scenario 1 represents the original inner-layer model before convexification as (16). Scenario 2 represents the model after convexification of the constraints related to the HVDC transmission systems. Scenario 3 represents the inner-layer model after convexification as (40). The table lists the SVSM results calculated by the corresponding model in each scenario when the RES output is 500 MW, 600 MW, 1000 MW, 1300 MW and 1400 MW, respectively. The inner-layer SVSM calculation models before and after convex relaxation are solved by the CONOPT and CPLEX solvers in the GAMS software, respectively. It can be seen that after convexification of scenarios 2 and 3, the errors of SVSM results are less than 0.01 with the SVSM of scenarios 1 as the benchmark. Thus, the SVSMs calculated by the inner-layer models before and after convexification are close, and the proposed convexification methods have high computational accuracy. The consumed CPU time of scenarios 2 and 3 are close with that of scenario 1, and they are all less than 0.25 s. Therefore, the inner-layer model after convexification can obtain the SVSM accurately, and the KKT condition can reflect the relationship between the SVSM and the control variables accurately. It is beneficial to the solution of the bi-level optimal SVSM enhancement control model for the application convex relaxation methods.

B. Analysis of Optimal SVSM Enhancement Control

According to (10), the minimum secure limit of SVSM $\lambda_{\min}$ is given to be 0.15. When the RES output is 500 MW or 1400 MW, the SVSM is less than 0.15, which does not meet the minimum secure limit requirement. At this time, the SVSM needs to be enhanced to a secure level through solving the proposed optimal SVSM enhancement control model. The SVSM calculated by the original model is compared with the SVSM results after the optimal SVSM enhancement control, and they are shown in Table III. It can be seen that after executing the optimal SVSM enhancement control schemes, the system SVSMs under these two RES outputs are bigger than 0.15 and have satisfied the given SVSM limit requirement, which demonstrates that the proposed optimal SVSM enhancement control model can obtain the control scheme that enhance the SVSM to satisfy the given secure operation limit requirement effectively.

In order to compare the influence of each control variable on the SVSM in the optimal SVSM enhancement control model, Tables IV–VI show the variation of each control variable. Tables IV and V show the comparison of generators' initial active power output $P_{\mathrm{G}i0}$ and terminal voltage' reference value $V_{\mathrm{G}i\mathrm{ref}}$ under different RES output scenarios before and after the optimal control. Table VI shows the comparison of the control parameters of DC converter stations' control modes under different RES output scenarios before and after the optimal control. From Table IV, it can be seen that $P_{\mathrm{G}i0}$ values after the optimal control have a slight decrease. This is because the small decrease of $P_{\mathrm{G}i0}$

TABLE III
COMPARISON OF SVSM VALUES BEFORE AND AFTER THE OPTIMAL CONTROL

P_{RE} /MW	Before control	After control
500	0.1451	0.1773
1400	0.1321	0.2336

TABLE IV
COMPARISON OF GENERATORS' ACTIVE POWER OUTPUT BEFORE AND AFTER THE OPTIMAL CONTROL

Bus	Before control	After control	
		P_{RE} = 500 MW	P_{RE} = 1400 MW
30	250	246	222
31	1000	958	937
32	650	649	649
33	632	628	624
34	508	503	501
35	650	649	647
36	560	556	559
37	540	523	526
38	830	827	818

TABLE V
COMPARISON OF GENERATORS' TERMINAL VOLTAGE BEFORE AND AFTER THE OPTIMAL CONTROL

Bus	Before control	After control	
		P_{RE} = 500 MW	P_{RE} = 1400 MW
30	1.047	1.047	1.047
31	1.030	1.032	1.040
32	0.983	0.983	0.983
33	0.997	1.010	1.037
34	1.012	1.012	1.012
35	1.049	1.050	1.050
36	1.063	1.063	1.063
37	1.027	1.036	1.045
38	1.026	1.027	1.048

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TABLE VI
COMPARISON OF CONTROL MODES' PARAMETERS OF LCC/VSC-HVDC CONVERTER STATIONS BEFORE AND AFTER THE OPTIMAL CONTROL

Control Variables	Before control	After control	
		P_{RE} = 500 MW	P_{RE} = 1400 MW
U_{d4ref}	1.24	1.21	1.21
I_{d4ref}	0.10	0.10	0.10
$\cos\gamma_{4ref}$	0.95	0.96	0.99
I_{d14ref}	0.30	0.30	0.30
$\cos\alpha_{14ref}$	0.95	0.96	0.99
k_{14ref}	3.50	3.50	3.79
V_{11ref}	0.90	0.90	0.93
V_{d6ref}	0.90	0.90	0.93
V_{d12ref}	0.90	0.90	0.93
Q_{d11ref}	-0.10	-0.13	-0.19
Q_{d6ref}	-0.80	-0.81	-0.84
Q_{d12ref}	-0.25	-0.26	-0.26
U_{d11ref}	2.00	2.00	2.00
U_{d6ref}	1.990	1.990	1.990
U_{d12ref}	1.994	1.994	1.994
P_{d6ref}	-1.90	-1.85	-1.62
P_{d12ref}	-1.50	-1.49	-1.34
k_{11ref}	-1.00	-0.91	-0.86

TABLE VII
COMPARISON OF CONTROL MODE SWITCHING OF LCC/VSC-HVDC CONVERTER STATIONS BEFORE AND AFTER THE OPTIMAL CONTROL

P_{RE} /MW	Converter station	Control Mode	
		Before control	After control
500	LCC at bus 14	I_d	I_d
	VSC at bus 11	Q_d, U_d	Q_d, U_d
1400	LCC at bus 14	I_d	k_{14}
	VSC at bus 11	Q_d, U_d	V_i, U_d

can increase the power growth margin of P_{Gi}. The maximum load growth that the system can bear will also increase, hence the system SVSM is enhanced. From Table V, it can be seen that $V_{Gi\mathrm{ref}}$ values after the optimal control have a small increase, which can reduce the network loss at the current operation state, and the maximum load growth that the power system can bear in the current operating state will increase, thus, the SVSM of the system can be effectively improved.

From Table VI, it can be seen that each control mode's parameter variation of DC converter stations is small. The control modes whose parameters are regulated in the optimal SVSM enhancement control are VDCOL control, constant U_d control, constant extinction angle control and constant rectifier's delay trigger angle control of LCC converter stations, and constant Q_d control, constant V_i control, constant P_d control and P_d-U_d droop control of VSC converter stations. These control parameter variations are beneficial to enhance the system SVSM.

Comparison of control modes of a LCC converter station at bus 14 and a VSC converter station at bus 11 before and after the optimal control is shown in Table VII. It can be seen that when $P_{RE} = 1400$ MW, the control mode of VSC rectifier station at bus 11 switches from constant Q_d control mode to constant V_i control mode with the load growth by adjusting the control parameters. As the load growth, the voltage at bus 11 V_{11} continues to decrease and it is close to the parameter $V_{11\mathrm{ref}}$ of the constant V_i control mode. At this time, after executing the optimal SVSM enhancement control scheme, the constant parameter $V_{11\mathrm{ref}}$ is increased, and V_{11} is easier to achieve $V_{11\mathrm{ref}}$. Thus, the control mode switching from constant Q_d control mode to constant V_i control mode is triggered, which is beneficial to the enhancement of SVSM. As shown in Table III, the increase of SVSM is 0.1015. When $P_{RE} = 500$ MW, by adjusting the control modes' parameters, the control mode switching of this VSC rectifier station is not triggered, and the increase of SVSM is small, only 0.0322. It can be seen that the control mode switching of the converter station is considered in the SVSM control, and its regulation can also help the system improve its static voltage stability.

In addition, it can also be seen that when $P_{RE} = 1400$ MW, the control mode switching of the LCC converter station at bus 14 is triggered from constant I_d control to VDCOL control. After executing the optimal SVSM enhancement control, $k_{14\mathrm{ref}}$ is increased from 3.50 to 3.79, which makes the intersection point of constant I_d control and VDCOL control closer to the initial operating point in the U_d-I_d coordinate system of Fig. 1. As the load growth, the control mode switching is easier to be triggered. When the control mode of bus 14 is switched into VDCOL control, U_{d14} and V_{14} decrease slower with the load growth due to the decrease of DC current, which helps the system enhance its SVSM.

TABLE VIII
COMPARISON OF SVSM VALUES AND CPU TIME IN DIFFERENT SCENARIOS

P_{RE} /MW	Before convexification	CPU time/s	After convexification	CPU time/s
(600, 200, 200)	0.1328	0.202	0.1353	0.219
(842, 300, 300)	0.1708	0.218	0.1768	0.225
(800, 250, 250)	0.1628	0.193	0.1692	0.223
(1000, 100, 100)	0.1450	0.226	0.1496	0.236

C. Analysis of Optimal SVSM Enhancement Control Results With Multiple RESs

In order to reflect the scenario of high penetration of RESs, the situation of multiple RESs integrated into the modified IEEE 39-bus system is analyzed. The RES connected at bus 9 is denoted as RES1, whose active power fluctuation interval of RES1 is [600, 1000] MW. Two RESs denoted as RES2 and RES3 are added in the system, and they are connected at bus 1 and bus 18, respectively. Their given active power fluctuation ranges are both [100, 300] MW. In order to demonstrate the effectiveness of the proposed convex approximation method of the inner-layer SVSM calculation model, comparison of the obtained SVSM results of 4 different RES output scenarios by solving the inner-layer SVSM calculation model before convexification (16) and the model after convexification (40) is shown in Table VIII. It can be seen that the errors of the SVSM results after convexification are still less than 0.01 compared to the SVSM results before convexification. Thus, after applying the convex approximation method of extended PFEs, the triangular convex hull relaxation method and other used convex relaxation methods to the inner-layer model, the errors of the obtained SVSM

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TABLE IX
COMPARISON OF SVSM VALUES BEFORE AND AFTER THE OPTIMAL CONTROL

P_{RE} /MW	Before control	After control
(600, 200, 200)	0.1328	0.1652
(1000, 100, 100)	0.1450	0.1744

TABLE X
COMPARISON OF SVSM, CONTROL COST AND CPU TIME OF TWO METHODS

P_{RE} /MW	SVSM before control	Proposed method			Sensitivity-based method		
		SVSM	control cost	CPU time/s	SVSM	control cost	CPU time/s
500 MW	0.1451	0.1773	270.31	0.728	0.1869	314.50	0.251
1400 MW	0.1321	0.2336	394.10	0.755	0.2540	453.39	0.263

are small under the high penetration of RESs. The consumed CPU time of the model after convexification is close with that of the model before convexification, and they are all less than 0.25 s. The results demonstrate that the proposed convexification methods can obtain the SVSM accurately.

The above RES output scenarios also include two insecure scenarios that the SVSM is less than 0.15, which does not meet the minimum secure limit requirement. Through solving the proposed optimal SVSM enhancement control model, the SVSM can be enhanced to be higher than 0.15. Comparison of SVSM results of multiple RESs before and after control are shown in Table IX. It can be seen that considering the uncertain fluctuations of three RESs, the system SVSM is easier to be lower than the minimum secure limit, which is required to execute the optimal SVSM enhancement control during operation. After executing the optimal SVSM enhancement control schemes, the system SVSMs under these two scenarios are both bigger than 0.15, which satisfy the given SVSM limit requirement. Thus, with multiple RESs penetrated into the power system, the proposed optimal SVSM enhancement control method can also effectively obtain the control scheme that can enhance the SVSM to satisfy the given secure limit requirement.

D. Comparison With Other Optimal SVSM Enhancement Control Methods

In order to highlight the advantages of the proposed optimal SVSM enhancement control method, the sensitivity-based optimal control method is applied to solve the proposed optimal SVSM enhancement control model (18) for comparison. The sensitivity-based control method uses an approximate linear expression to describe the relationship of the SVSM with respect to control variables, and this linear expression is used to replace the inner-layer SVSM calculation model (16)–(17) in the optimal SVSM enhancement control model (18). For the system with an RES at bus 9, comparison of the solution results of the two methods is shown in Table X. It can be seen that the SVSM obtained by the sensitivity-based method is higher than the proposed method, but the control cost is also higher. This is because of the approximate linear relationship of SVSM and control variables, which result in the excessive adjustment of control parameters. In addition, from the perspective of CPU time, the sensitivity-based method consumes less CPU time than the proposed method, and it is more efficient. However, the sensitivity-based control method is required to consume lots of time to calculate the sensitivities in advanced.

V. CONCLUSION

In this study, an optimal SVSM enhancement control model of AC/DC power system with RESs is proposed. In this model, the control mode switching constraints of DC converter stations described by multiple complementary equations are added. By adjusting the generators' initial active power output and terminal voltage and the parameters of various control modes of DC converter stations, the system SVSM under the uncertain RES output is enhanced. By convex approximation of the extended PFEs, and the triangular convex hull relaxation of the control mode switching constraints of DC converter stations, the inner-layer SVSM calculation model is transformed into its KKT conditions, then the proposed bi-layer optimal SVSM enhancement control model is transformed into a single-layer optimization model for direct solution. Case study on the modified IEEE-39 bus system shows that the proposed convex approximation and the triangular convex hull relaxation method have high computational accuracy. The optimal control scheme obtained by solving the proposed model can effectively enhance SVSM to be bigger than the secure operation requirements. Taking the control mode switching of the DC converter stations with the load growth into account in the optimal SVSM enhancement control is beneficial to improve the SVSM of the AC/DC power system.

The proposed method can be applied to guide the operation of actual AC/DC power systems to execute the preventive control measures to increase the SVSM of current operating point to meet the secure operation requirements. The preventive control measures include regulating the generators' initial active power output and terminal voltage and DC stations' control mode parameters. It can effectively avoid the occurrence of voltage instability and power outage cases in the system operation.

This paper only focuses on the optimal SVSM enhancement control problem under normal operation of an AC/DC power system. However, the influence of DC converter stations' control mode switching on SVSM can also be considered in the optimal SVSM enhancement control problem under multiple *N*-1 contingencies. How to solve this problem is a possible direction of future work.

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